Case-disjoint benchmark of surrogate models for flow and thermal fields in supercritical CO_2 natural-circulation loops

Abstract

Surrogate models for supercritical carbon dioxide (sCO_2) natural-circulation loops (NCLs) can replace day-long steady-state conjugate computational fluid dynamics (CFD) simulations with millisecond-scale field predictions, yet most published evaluations rely on random node-level train–test splits or single pooled metrics that mask generalization failure across unseen operating conditions. Here we present a case-disjoint surrogate benchmark built from 42 steady-state RANS simulations solved with a k–ε family turbulence model (k and ε are exported) and a Span–Wagner equation of state verified against `CoolProp` (median 0.0008%) across three heater–cooler configurations (`HHHC`, `HHVC`, and `VHVC`) and two loop scales. Across three random seeds, we benchmark 17 surrogate models (plus an input-concatenation conditioning ablation) spanning tree ensembles, Fourier-coordinate neural networks, physics-informed neural networks (PINNs), DeepONet, Latent Neural Operator, Transolver-style attention, multi-scale graph networks, Fourier neural operators (`F-FNO`, `U-FNO`), and proper orthogonal decomposition (`POD`) reduced-order models under five-fold stratified case-disjoint cross-validation and leave-one-group-out extrapolation. We show that random node splitting understates five-fold case-disjoint error across the four nonlinear tree baselines by roughly an order of magnitude on thermodynamic state fields (6.78×–10.37× on density, 11.29×–87.54× on gauge pressure, and 5.62×–10.24× on temperature in case-macro MAE) because high-capacity models memorize case-specific operating levels from adjacent mesh nodes. We also show that separating closed-loop gauge pressure into its case-level thermal-pressurization offset and centered spatial residual prevents small within-case hydrostatic variations from distorting model rankings. Across the eight exported CFD fields, no single architecture wins universally: among cross-configuration pooled models, `DeepONet` and `PINN continuity` lead on density and temperature, `Extra Trees` best predicts case-level gauge pressure, and multiplicative feature-modulated coordinates (`Domain-informed FiLM-MLP`) achieve the lowest error across both turbulence fields (k, ε) and all three velocity components (u, v, w), while within-group `POD-RBF` yields the sharpest temperature and turbulence reconstructions when interpolating on a fixed loop mesh.

Keywords: supercritical CO_2, natural-circulation loop, CFD surrogate benchmark, case-disjoint validation, neural operators, proper orthogonal decomposition

1 Introduction

Natural-circulation loops (NCLs) operating with supercritical fluids remove heat without mechanical pumps by balancing buoyancy forces between heated and cooled sections against frictional and local losses [1, 2, 3]. Supercritical carbon dioxide (sCO_2, critical point $T_c = 304.13$ K $= 30.98°$C, $p_c = 7.3773$ MPa $= 73.773$ bar) is widely used as a scaled simulant fluid for supercritical water reactor (SCWR) thermal-hydraulics and compact power cycles because its lower critical point permits experiments and conjugate CFD studies near the pseudocritical locus [4, 5, 6]. Near the pseudocritical temperature, steep variations in density ρ, specific heat c_p, viscosity μ, and thermal conductivity k strongly couple buoyancy gradients with turbulence production and heat transfer [7, 8, 9].

Because resolving conjugate heat transfer between the primary sCO_2 pipe, the solid metallic wall, and the secondary cooling-water jacket requires fine near-wall meshes and real-gas property evaluations, each steady-state RANS simulation in our dataset requires roughly 24 hours of compute time. Data-driven spatial surrogates—including gradient-boosted trees [10, 11, 12], coordinate neural networks [13, 14, 15], neural operators [16, 17, 18, 19], and proper orthogonal decomposition (POD) reduced-order models [20, 21]—provide

millisecond-scale predictions once trained. However, three practical issues complicate surrogate benchmarking on closed natural-circulation loops: (i) random node-level train–test splits leak case-specific operating levels across neighboring mesh nodes; (ii) pooled metrics such as R^2 conflate between-case operating-level shifts with within-case spatial profile accuracy, while naive within-case normalization breaks down for gauge pressure in closed loops; and (iii) slender circular pipes ($D_i/L \sim 10^{-2}$) and varying heater–cooler orientations create different inductive-bias requirements for thermodynamic state fields versus velocity and turbulence fields.

To address these issues, this paper contributes: (1) a controlled comparison of 80/20 random-node splitting against five-fold case-disjoint cross-validation across four nonlinear tree models and two linear/constant controls; (2) a physical characterization and two-part evaluation protocol for closed-loop gauge pressure; and (3) a three-seed benchmark of 18 surrogate model variants across eight CFD fields under both five-fold case-disjoint interpolation and leave-one-group-out geometric extrapolation.

2 Physical Domain and sCO_2 Natural-Circulation Loop Dataset

2.1 Loop Geometry, CFD Solver Configuration, and Primary Fluid Domain

The dataset consists of 42 converged steady-state RANS conjugate heat-transfer simulations of closed rectangular sCO_2 natural-circulation loops solved in ANSYS Fluent and exported on cross-sectional planes via ANSYS CFD-Post. All simulations use a k–ε family turbulence model (k and ε are exported) and a Span–Wagner equation of state verified against `CoolProp` [4, 22] (median relative density discrepancy 0.0008%) for the primary sCO_2 fluid.

The 42 cases cover three heater–cooler configurations—horizontal heater/horizontal cooler (`HHHC`), horizontal heater/vertical cooler (`HHVC`), and vertical heater/vertical cooler (`VHVC`)—across two loop scales (Table 1): 29 small-loop cases ($L_x \times L_y = 0.8936 \times 1.0000$ m, $D_i = 15.00$ mm) and 13 large-loop `VHVC` cases ($L_x \times L_y = 3.0100 \times 4.0800$ m, $D_i = 13.84$ mm). Complete radial dimensions of the stainless-steel wall and cooling-water jacket, axial heater/cooler spans, and sampling-plane counts (97 planes in small loops; 142 planes in the large loop) are given in Supplementary Section S1.

Each simulation is defined by four boundary parameters encoded in its case name `{CONFIG}_P{Q}_{p}_bar_{mdot}kgs_T_{Tin}`: heater power $Q \in [10, 2500]$ W, operating pressure $p_{\text{op}} \in \{80, 85, 90, 100\}$ bar, cooling-water mass flow rate $\dot{m} \in [0.010, 0.150]$ kg s^{-1}, and cooling-water inlet temperature $T_{\text{in}} \in \{283.5, 300, 305, 315\}$ K (Fig. 1). At each node (x, y, z), eight CFD target fields are exported: density ρ (kg m^{-3}), gauge pressure p_{gauge} (Pa), temperature T (°C), turbulence eddy dissipation ε (m^2 s^{-3}), turbulence kinetic energy k (m^2 s^{-2}), and velocity components (u, v, w) (m s^{-1}). Removing solid wall nodes (`null` velocity/turbulence, $\rho \approx 7970$ kg m^{-3}) and outer cooling-water jacket nodes ($\rho > 950$ kg m^{-3}) and deduplicating intersecting plane coordinates yields 5,451,341 unique primary sCO_2 nodes ($\rho \in [153.2, 892.5]$ kg m^{-3}; Fig. 2).

Table 1: Summary of the 42 steady-state CFD cases by heater–cooler configuration and loop scale. Full radial and plane-sampling details are in Supplementary Section S1.

Config.	Scale	Cases	$L_x \times L_y$ (m)	D_i (mm)	Q (W)	p_{op} (bar)	$\dot{m}$ (kg s^{-1})	sCO_2 nodes/case
`HHHC`	Small	20	0.894×1.000	15.00	1000–2000	80–100	0.050–0.150	97,264
`HHVC`	Small	5	0.894×1.000	15.00	10–50	90	0.050	89,771
`VHVC`	Small	4	0.894×1.000	15.00	10–40	90	0.050	89,647
`VHVC`	Large	13	3.010×4.080	13.84	250–2500	85	0.010–0.035	207,586
Total	**4 groups**	**42**	–	–	**10–2500**	**80–100**	**0.010–0.150**	**5,451,341 total**

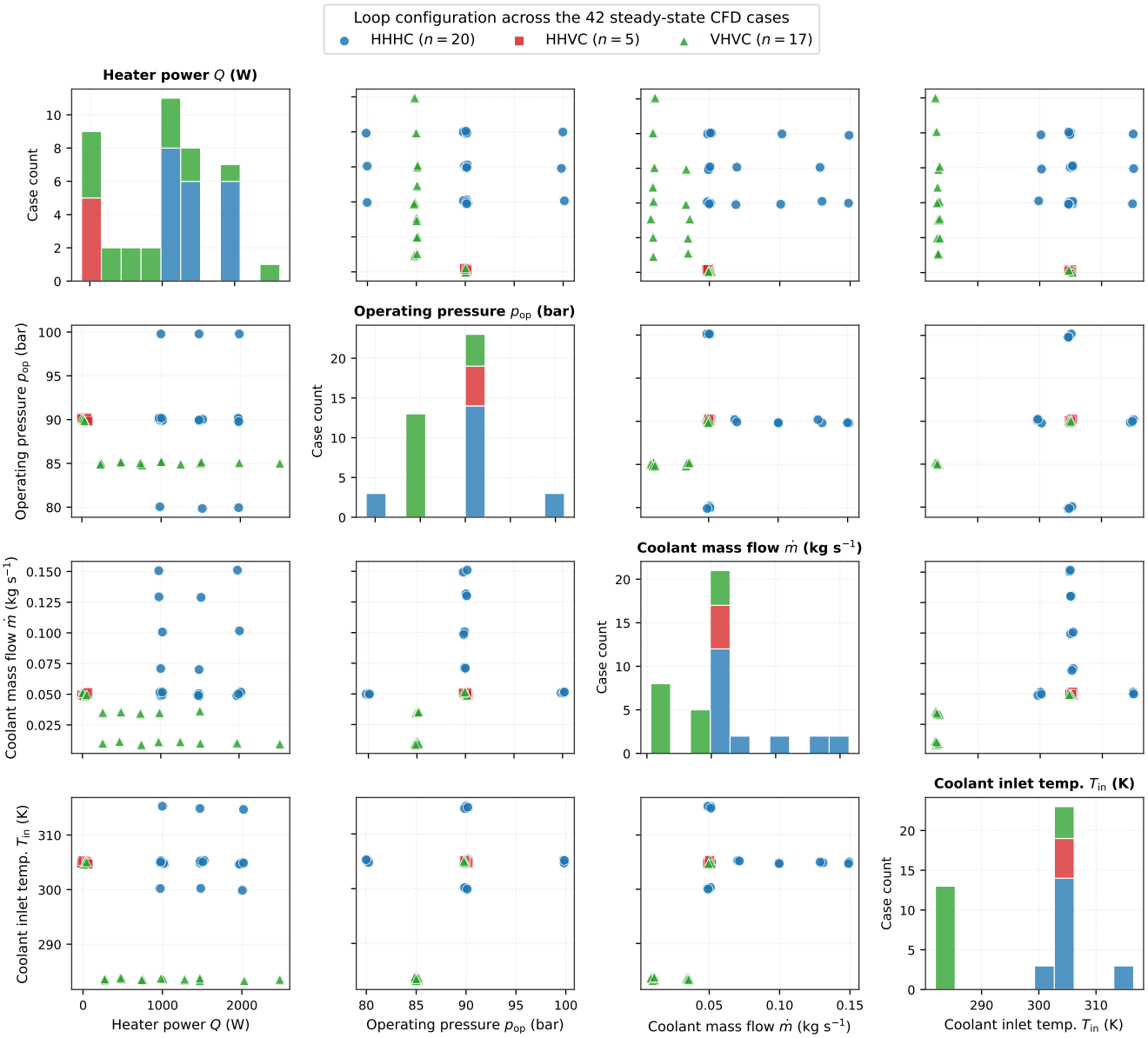


Figure 1: Operating-condition distribution across the 42 steady-state CFD cases by heater–cooler configuration (`HHHC`, `HHVC`, `VHVC`). Diagonal panels show stacked case-count histograms for Q, p_{op}, $\dot{m}$, and T_{in}.

2.2 Behavior of Exported Gauge Pressure Across Cases

In all 42 simulations, the local absolute pressure used to evaluate CO_2 properties is the sum of the operating pressure p_{op} and the exported gauge pressure $p_{\mathrm{gauge}}(\mathbf{x})$:

$$p_{\mathrm{abs}}(\mathbf{x}) = p_{\mathrm{op}} + p_{\mathrm{gauge}}(\mathbf{x}). \tag{1}$$

Evaluating $(T(\mathbf{x}), p_{\mathrm{abs}}(\mathbf{x}))$ in `CoolProp` [4, 22] matches the exported CFD density $\rho(\mathbf{x})$ across all 42 cases to a median relative difference of 0.0008% (0.00004%–0.0015%). Within any single case c, p_{gauge} varies very little around the loop because the solver subtracts a constant reference hydrostatic head: the within-case standard deviation $\sigma_c(p_{\mathrm{gauge}})$ is only 0.15–0.39 kPa in the small loops and 1.45–8.79 kPa in the large loop (42-case mean $\bar{\sigma}_c = 1.51$ kPa).

Across different cases, however, the case-mean gauge pressure $\bar{p}_c$ spans 1.42 kPa to 1.763 MPa (Fig. 3):

- **Large loop vs. small loops:** In the 13 large-loop `VHVC` cases, the reference gauge pressure is pinned near 0 Pa at the top of the loop ($y = 4.08$ m), keeping $\bar{p}_c \in [1.42, 8.77]$ kPa (Fig. 3a). In the 29

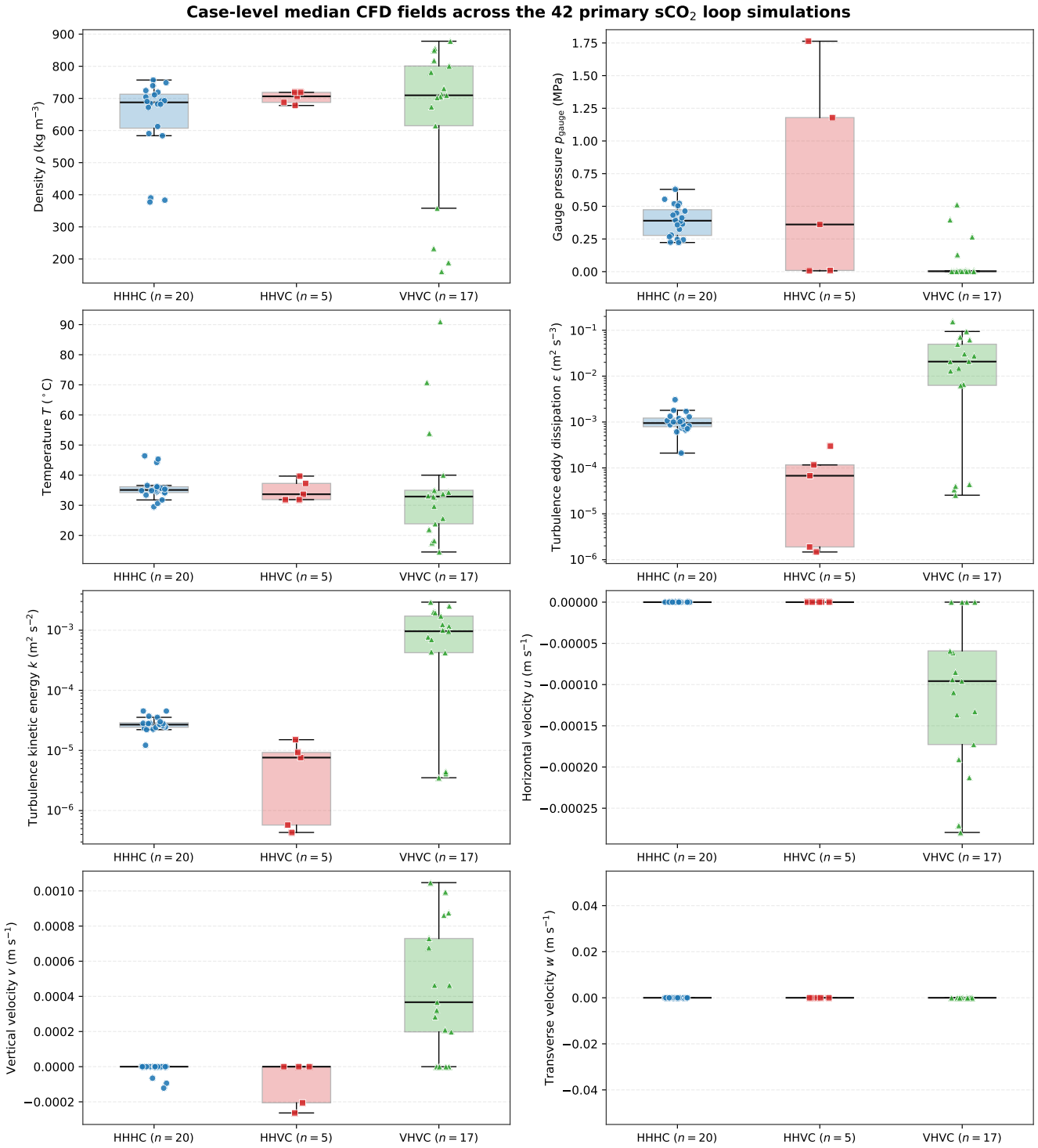


Figure 2: Case-level median distributions of the eight exported CFD fields across the 42 primary sCO_2 loop simulations (5, 451, 341 nodes), grouped by configuration (`HHHC`, `HHVC`, `VHVC`).

small-loop cases (`HHHC`, `HHVC`, and small `VHVC`), the closed fluid domain has no pressure-outlet pin, so $\bar{p}_c$ rises with the loop-mean temperature $\bar{T}_c$.

- **Empirical pressure–temperature slope vs. `CoolProp`:** Across the 17 small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, a linear fit of $\bar{p}_c$ versus $\bar{T}_c$ gives an *empirical slope* of 0.214 MPa K^{-1} (0.235 MPa K^{-1}, or $\approx$ 0.23 MPa K^{-1}, across the five `HHVC` cases; Fig. 3b). By comparison, the theoretical constant-density derivative $(\partial p/\partial T)_\rho$ computed in `CoolProp` at each case's mean den-

sity and temperature is 0.315–0.365 MPa K^{-1} (median 0.338 MPa K^{-1}, or $\approx$ 0.35 MPa K^{-1} near $\rho = 710$ kg m^{-3} and $T = 305$–308 K; 0.133–0.417 MPa K^{-1} across all 29 small-loop cases in Supplementary Table S0b). Testing the fixed-mass hypothesis via the plane-weighted mean CO_2 density $\bar{\rho}_{\mathrm{plane},c}$ across cases at the same $(p_{\mathrm{op}}, T_{\mathrm{in}})$ shows that $\bar{\rho}_{\mathrm{plane},c}$ is not strictly constant: at 90 bar, 305 K it spans 671.6–704.0 kg m^{-3} (SD = 9.79 kg m^{-3}) in `HHHC`, 681.2–717.8 kg m^{-3} (SD = 16.66 kg m^{-3}) in `HHVC`, and 696.8–712.1 kg m^{-3} (SD = 6.62 kg m^{-3}) in small `VHVC` (and 161.5–860.8 kg m^{-3}, SD = 270.83 kg m^{-3} in top-pinned `VHVC/large`). Because $\bar{\rho}_{\mathrm{plane},c}$ decreases slightly in warmer cases ($\mathrm{d}\bar{\rho}/\mathrm{d}\bar{T} < 0$), the relation $\mathrm{d}\bar{p} \approx (\partial p/\partial T)_\rho \, \mathrm{d}\bar{T} + (\partial p/\partial \rho)_T \, \mathrm{d}\bar{\rho}$ yields an empirical slope lower than the constant-density `CoolProp` value.

- **Near-stagnant steady solutions in `HHVC` at 20 W and 30 W:** In the five `HHVC` cases at 90 bar and 305 K, the $Q = 20$ W and 30 W runs (`HHVC_P20_90_bar_0.05kgs_T_305` and `HHVC_P30_90_bar_0.05kgs_T_305`) are near-stagnant steady solutions (mean speed $\overline{|\mathbf{u}|} = 0.9$–1.1 mm s^{-1}, $\bar{T}_c = 31.92$–31.94°C, $\bar{p}_c = 6.1$–9.2 kPa), whereas the 10 W, 40 W, and 50 W runs have sustained buoyant circulation ($\overline{|\mathbf{u}|} = 0.044$–0.107 m s^{-1}, $\bar{T}_c = 33.58$–39.36°C, $\bar{p}_c = 0.361$–1.763 MPa; Fig. 3b).

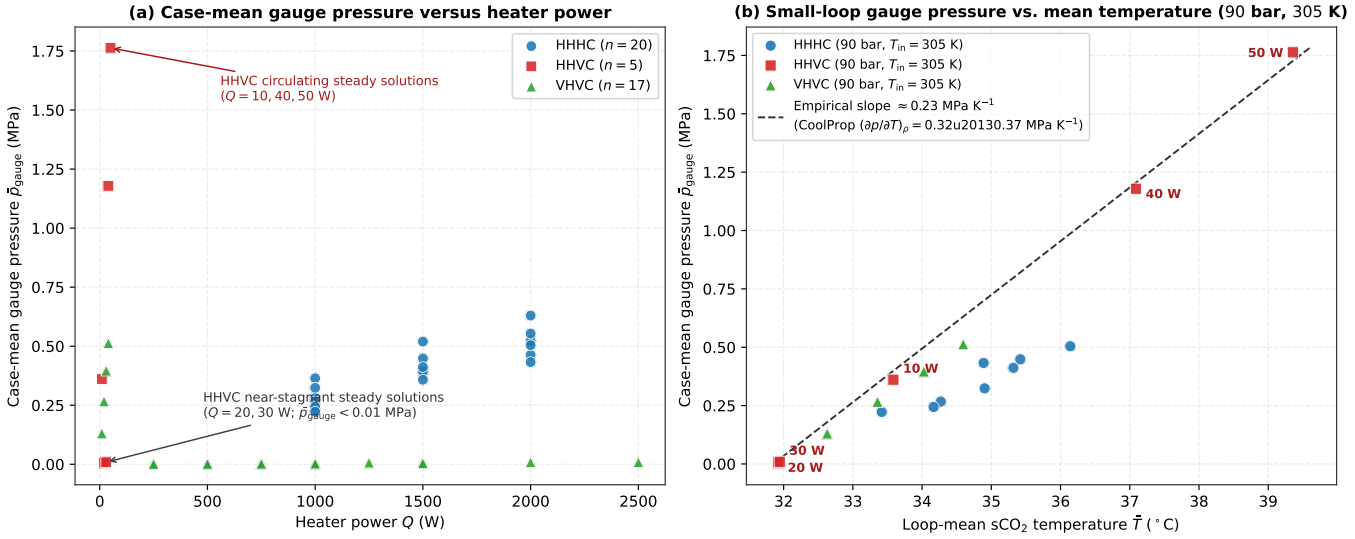


Figure 3: Case-level gauge-pressure behavior across the 42 sCO_2 simulations. (a) Case-mean gauge pressure $\bar{p}_{\mathrm{gauge}}$ versus heater power Q. (b) In the 17 small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, $\bar{p}_{\mathrm{gauge}}$ follows an empirical slope of 0.214 MPa K^{-1} (0.235 MPa K^{-1} across `HHVC`) versus loop-mean temperature $\bar{T}_c$, compared against the `CoolProp` constant-density derivative $(\partial p/\partial T)_\rho = 0.315$–0.365 MPa K^{-1} (median 0.338 MPa K^{-1}).

3 Benchmark Methodology, Models, and Evaluation Protocol

3.1 Validation Protocols and Evaluation Metrics

Every model predicts the eight target fields $\mathbf{y}(\mathbf{x}, \mathbf{c}) \in \mathbb{R}^8$ at coordinates $\mathbf{x} = (x, y, z)$ given the case descriptor $\mathbf{c} = (Q, p_{\mathrm{op}}, \dot{m}, T_{\mathrm{in}}, L_x, L_y, D_i, \text{one-hot config})$. We evaluate three validation protocols:

1. **Five-fold stratified case-disjoint cross-validation:** For each seed $s \in \{2026, 2027, 2028\}$, the 42 cases are split into five folds (9, 9, 8, 8, 8 cases) stratified by the four groups (`HHHC/small`, `HHVC/small`, `VHVC/small`, `VHVC/large`). Each seed independently re-randomizes the five-fold case partition, the four validation cases per fold, the training node subsample (30,000 nodes/case in GPU 3-seed CV; 20,000 in the CPU split test), and model initialization. Held-out test cases are evaluated on 100% of their mesh nodes (5,451,341 out-of-fold predictions per seed).

2. **Within-group mesh-aligned cross-validation:** Mesh-aligned models (`POD-RBF`, `POD-GPR`, `Nearest case`) require a shared mesh ordering and therefore train and predict strictly within the matching configuration/scale group of each five-fold split.

3. **Leave-one-group-out extrapolation:** All cases in one structural group (`HHHC/small`, $n = 20$; `HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$; or `VHVC/large`, $n = 13$) are held out while the 16 cross-configuration pooled models train on the remaining three groups (16 models×8 fields = 128 model–field evaluations per group).

For each held-out case c (N_c nodes, truth $y_{c,i}$, prediction $\hat{y}_{c,i}$, within-case standard deviation σ_c), we compute case-level MAE, RMSE, case-normalized MAE ($\mathrm{nMAE}_c = \mathrm{MAE}_c/\sigma_c$), and node-pooled R^2_{pooled}:

$$\mathrm{MAE}_c = \frac{1}{N_c}\sum_{i=1}^{N_c}|\hat{y}_{c,i} - y_{c,i}|, \quad \mathrm{RMSE}_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}(\hat{y}_{c,i} - y_{c,i})^2}, \quad R^2_{\mathrm{pooled}} = 1 - \frac{\sum_{c,i}(\hat{y}_{c,i} - y_{c,i})^2}{\sum_{c,i}(y_{c,i} - \bar{y}_{\mathrm{all}})^2}. \tag{2}$$

Across the 42 cases, we report case-macro MAE, case-macro RMSE, median case nMAE, and R^2_{pooled}. Because $\bar{\sigma}_c(p_{\mathrm{gauge}}) = 1.51$ kPa is small relative to between-case pressure shifts (Section 2.2), we suppress raw nMAE_c for p_{gauge} and evaluate the case-mean level error $e_{\mathrm{level},c}$ and centered spatial residual $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$ separately:

$$e_{\mathrm{level},c} = |\hat{\bar{p}}_c - \bar{p}_c|, \quad \mathrm{RMSE}_{\mathrm{res},c} = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}\left(\hat{\tilde{p}}_{c,i} - \tilde{p}_{c,i}\right)^2}, \quad \mathrm{nRMSE}_{\mathrm{res},c} = \frac{\mathrm{RMSE}_{\mathrm{res},c}}{\max(\sigma_c, 1\ \mathrm{kPa})}. \tag{3}$$

3.2 Benchmark Model Suite and Verified Hyperparameters

Table 2 summarizes all evaluated models, their conditioning mechanisms, and their verified hyperparameters from `notebooks/kaggle_clean_case_cv_predictions.ipynb` and `scripts/cpu_clean_benchmarks.py`. To isolate multiplicative feature modulation in coordinate networks, `Domain-informed FiLM-MLP` modulates each hidden block by $\mathbf{h}^{(\ell)} \odot (1 + \boldsymbol{\gamma}^{(\ell)}(\mathbf{c})) + \boldsymbol{\beta}^{(\ell)}(\mathbf{c})$ [15], whereas its matched ablation `Domain-informed Concat-MLP` concatenates $\mathbf{c}$ at the input layer.

4 Results and Discussion

4.1 Optimism of Node-Random Splits Versus Case-Disjoint Validation

Table 3 and Figure 4 compare 80/20 random node splitting against five-fold case-disjoint cross-validation on the 5,451,341 primary sCO_2 nodes (seed `2026`). Whereas the two low-capacity controls (`Mean baseline` and `Ridge`) show error ratios near 1.0× because they cannot memorize local mesh neighborhoods, the four nonlinear tree models (`Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`) experience large error inflation under case-disjoint evaluation on thermodynamic state fields—ranging from 6.78 × –10.37× on density, 11.29 × –87.54× on gauge pressure, and 5.62 × –10.24× on temperature in case-macro MAE (3.34 × –4.05× and 2.95 × –4.39× in median case nMAE on ρ and T)—because random node splits allow high-capacity trees to interpolate case-specific operating levels from adjacent nodes of the same simulation.

4.2 Three-Seed Case-Disjoint Benchmark Across All Eight CFD Fields

Figure 5 shows the three-seed (`2026`, `2027`, `2028`) case-macro MAE of all 18 models relative to the `Mean baseline`, and Table 4 reports the detailed metrics of the top-performing pooled and within-group models for each field (complete 19-model tables are in Supplementary Tables S1–S8).

Four main findings emerge from Table 4 and Figure 5:

Table 2: Evaluated surrogate models, spatial representations, and verified hyperparameters.

Model	Family	Conditioning representation & verified hyperparameters
`Mean baseline`	Constant	Predicts training-fold node mean per target field.
`Ridge`	Linear	ℓ_2 linear regression ($\alpha = 10.0$) on standardized $[\mathbf{x}, \mathbf{c}]$.
`Random Forest`	Bagged trees	Multi-output CART on $[\mathbf{x}, \mathbf{c}]$: 48 trees (80 in CPU split test), max depth 20, max leaf nodes 4096 (2048 CPU), min leaf 5, max features 0.8, bootstrap 0.3.
`Extra Trees`	Random trees	Multi-output randomized trees on $[\mathbf{x}, \mathbf{c}]$: 64 trees (96 CPU), depth 20, max leaf 4096.
`XGBoost`	Boosted trees	Per-target `hist` trees: 300 trees (220 CPU), depth 8 (6 random-node CPU), lr 0.08, subsample 0.85, colsample 0.85, $\lambda = 2.0$.
`CatBoost`	Oblivious trees	Multi-output symmetric trees (`MultiRMSE`): 350 iterations, depth 8, lr 0.08, $\ell_2 = 3.0$.
`DeepONet`	Operator net	4-layer branch MLP on $\mathbf{c}$ + 4-layer trunk MLP on $\mathbf{x}$ (width 256, latent 128), MSE loss, AdamW (lr $= 5 \times 10^{-4}$), batch 8192 (8 min cap/fold).
`PINN data-only`	Coord. MLP	5-layer Fourier-coordinate MLP ($B = 4$ bands, width 256, GELU + LayerNorm), MSE loss.
`PINN continuity`	Physics MLP	Identical to `PINN data-only` + $\lambda_{\text{cont}} = 0.05 \|\nabla \cdot (\hat{\rho}\hat{\mathbf{u}})\|_2^2$ on 1024 collocation nodes/step.
`Domain-informed FiLM-MLP`	Modulated MLP	5-block residual MLP (width 256, $B = 4$ Fourier bands + leg centerline/arc-length features), FiLM modulation by $\mathbf{c}$, Huber loss ($\delta = 1.0$), AdamW (lr $= 5 \times 10^{-4}$, batch 8192).
`Domain-informed Concat-MLP`	Ablation MLP	Identical architecture and training to `FiLM-MLP`, but concatenates $\mathbf{c}$ at the input layer.
`Latent Neural Operator`	Latent attn.	Cross-attention from Fourier coordinates ($B = 4$) to $M = 24$ latent tokens (width 128, 4 heads, branch/query depth 3, decoder depth 2), Huber loss.
`Transolver-style`	Slice attn.	4-block physics-attention network (width 128, 4 heads, $S = 32$ slices), 8192 context / 4096 query nodes per step, AdamW (lr $= 3 \times 10^{-4}$, 2200 steps).
`GAMS-GNN-style`	Multiscale GNN	4-block message passing (width 96) across $k = 6$ fine kNN graph (8192 nodes) and $(20, 20, 6)$ / $(10, 10, 3)$ voxel graphs, AdamW (lr $= 2 \times 10^{-4}$, 1600 steps).
`F-FNO`	Spectral FNO	4 factorized 1D spectral blocks (width 24, 12 modes) on $64 \times 64 \times 16$ voxel grid, AdamW (lr $= 10^{-3}$, 2000 steps).
`U-FNO3D` (`U-FNO`)	U-Net FNO	3 spectral blocks + 3D U-Net branch (width 24, 12 modes) on $64 \times 64 \times 16$ voxel grid.
`POD-RBF` / `POD-GPR`	Aligned ROM	Within-group SVD (≤ 12 modes, 99.99% energy) + Gaussian RBF (LOO-CV over $\ell \in [0.25, 8], \alpha \in [10^{-6}, 1]$) or GPR ($\ell = 1, \alpha = 10^{-4}$).
`Nearest case`	Aligned lookup	Copies the closest training case in standardized operating-parameter space within the same group.

1. **State fields (ρ, T) vs. kinematic/turbulence fields (u, v, w, k, ε):** On density and temperature, pooled models reduce case-macro MAE by 73.3%–80.0% (ρ) and 60.0% (T) and reach high pooled R^2 (0.862–0.936 and 0.736), but narrow within-case standard deviations in the small loops ($\sigma_c(T) \approx$ 0.25–2.5°C) keep median case nMAE at 0.837 (`DeepONet` on ρ) and 1.350 (`PINN continuity` on T; reduced to 0.974 by within-group `POD-RBF`). Conversely, velocity and turbulence fields have near-zero case-mean offsets and strong geometry-locked spatial profiles; here `Domain-informed FiLM-MLP` achieves median case nMAE of 0.162 (u), 0.175 (v), 0.192 (ε), and 0.242 (k).

2. **FiLM modulation vs. input concatenation:** Relative to `Domain-informed Concat-MLP`, multiplicative FiLM conditioning reduces median case nMAE on six of the seven spatially varying fields: T (1.467 vs. 1.758), ε (0.192 vs. 0.226), k (0.242 vs. 0.273), u (0.162 vs. 0.224), v (0.175 vs. 0.230), and w (0.272 vs. 0.318), whereas concatenation yields lower case-macro MAE on density (26.3 vs. 31.3 kg m^{-3}).

3. **Gauge-pressure level vs. spatial residual:** Total gauge-pressure MAE (83.2 kPa for `Extra Trees`, a 67.3% reduction below the `Mean baseline`) is dominated by the case-level offset $e_{\text{level},c}$

Table 3: Error inflation ratios (five-fold case-disjoint error divided by 80/20 random-node error) across the four nonlinear tree models (`Random Forest`, `Extra Trees`, `XGBoost`, `CatBoost`), compared against the low-capacity `Mean baseline` and `Ridge` controls.

Target field	**4 nonlinear trees (Macro MAE ratio)**	**4 nonlinear trees (Median nMAE ratio)**	`Mean` control (MAE / nMAE)	`Ridge` control (MAE / nMAE)
Density ρ	**6.78–10.37**×	**3.34–4.05**×	0.96× / 1.03×	1.26× / 1.24×
Gauge pressure p_{gauge}	**11.29–87.54**×	n/a[†]	0.95× / n/a	1.09× / n/a
Temperature T	**5.62–10.24**×	**2.95–4.39**×	0.98× / 0.96×	1.25× / 1.26×
TED ε	1.05–1.39×	0.97–1.02×	1.00× / 0.97×	1.01× / 1.01×
TKE k	1.41–2.99×	0.99–1.11×	0.99× / 1.02×	1.10× / 1.02×
Velocity u	1.80–2.63×	1.34–1.61×	1.00× / 1.00×	1.00× / 1.00×
Velocity v	1.57–2.34×	1.10–1.29×	1.00× / 1.00×	1.00× / 1.00×
Velocity w	0.96 × –1.02×	0.97–1.02×	1.00× / 1.00×	1.00× / 1.00×

[†] Within-case nMAE is suppressed for gauge pressure because $\bar{\sigma}_c = 1.51$ kPa reflects only the within-case vertical pressure variation.

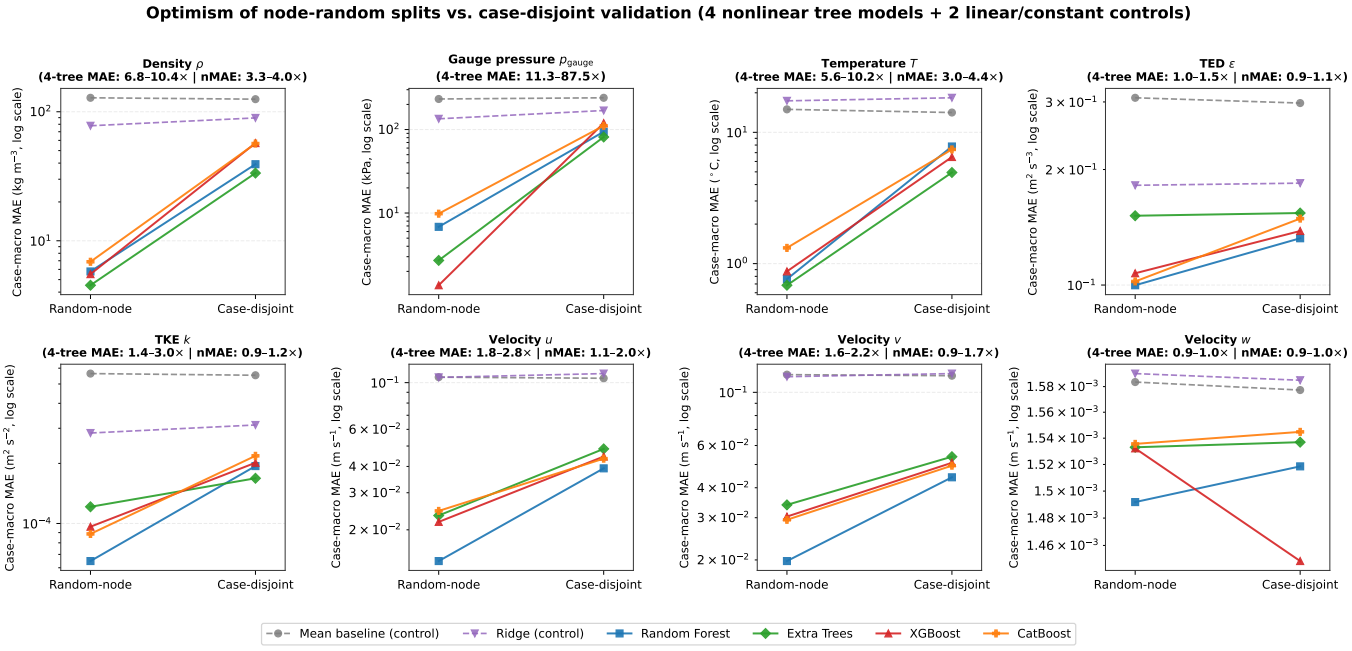


Figure 4: Case-macro MAE (log scale) under 80/20 random-node splits versus five-fold case-disjoint validation across the four nonlinear tree baselines (`RF`, `ET`, `XGB`, `CB`) and two controls (`Mean`, `Ridge`).

(82.9 kPa). On the centered spatial residual $\tilde{p}_{c,i}$ (Supplementary Table S8), mesh-aligned `Nearest case` ($\text{nRMSE}_{\text{res}} = 0.0122$), `POD-GPR` (0.0457), `POD-RBF` (0.0483), and `XGBoost` (0.151) preserve the smooth hydrostatic profile, whereas coordinate networks exhibit higher residual roughness ($\text{nRMSE}_{\text{res}} = 2.91$–$24.3$).

4. **Voxelized FNOs on slender pipe loops:** `F-FNO` and `U-FNO3D` rank near the bottom of the neural models (Fig. 5) because voxelizing thin pipes ($D_i/L < 1.7\%$) onto a $64 \times 64 \times 16$ Cartesian grid places only 1–2 cells across the pipe diameter, blurring radial boundary-layer gradients.

Exploratory statistical comparisons: Treating nonparametric Friedman tests and Holm-adjusted two-sided Wilcoxon signed-rank tests across the 42 cases (three-seed mean five-fold errors; Supplementary Table S10) as exploratory, overall model differences are significant on all eight fields ($p < 0.001$). Against

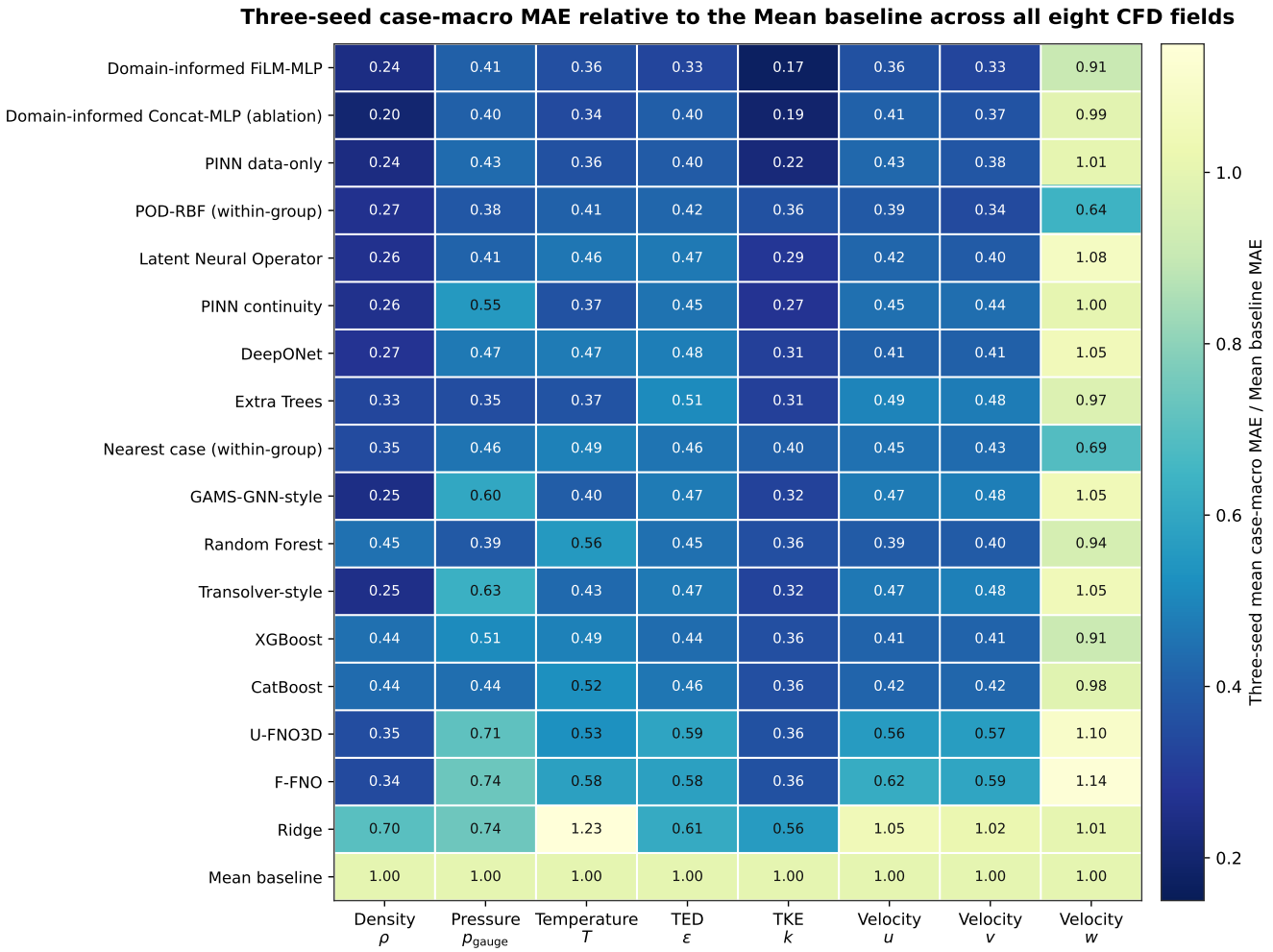


Figure 5: Three-seed mean case-macro MAE relative to the `Mean baseline` across all eight CFD fields, ordered by geometric-mean relative error.

`Domain-informed FiLM-MLP` ($\alpha = 0.05$): (i) on ε and k, `FiLM-MLP` ranks first (mean rank 4.62), significantly outperforming all 15 pooled baselines on ε and 14 of 15 on k (all except `Concat-MLP`, $p_{\text{Holm}} = 0.190$), while not differing significantly from within-group `POD-RBF`, `POD-GPR`, and `Nearest case` ($p_{\text{Holm}} \geq 0.190$); (ii) on u and v, `FiLM-MLP` ranks first (5.83 and 5.26), significantly outperforming `Concat-MLP`, both `PINNs`, `LNO`, `Transolver-style`, `GAMS-GNN-style`, both `FNOs`, `Ridge`, and `Mean baseline` ($p_{\text{Holm}} < 0.05$, plus `DeepONet` and `Extra Trees` on v), while not differing significantly from the remaining tree ensembles or within-group aligned models ($p_{\text{Holm}} \geq 0.052$); (iii) on w, `FiLM-MLP` is significantly outperformed by within-group `POD-RBF`, `POD-GPR`, and `Nearest case`, does not differ significantly from `PINN continuity`, tree ensembles, `Ridge`, or `Mean baseline` ($p_{\text{Holm}} \geq 0.096$), and significantly outperforms all other neural/operator baselines ($p_{\text{Holm}} < 0.05$); and (iv) on ρ and T, `FiLM-MLP` is significantly outperformed by `POD-RBF` on T, does not differ significantly from `DeepONet`, `Concat-MLP`, `PINNs`, `Transolver-style`, `GAMS-GNN-style`, or `Extra Trees` ($p_{\text{Holm}} \geq 0.170$), and significantly outperforms `F-FNO`, `U-FNO3D`, `Ridge`, and `Mean baseline` (plus `RF`, `XGB`, `CB`, and `POD-GPR` on ρ, and `LNO` on T).

4.3 Spatial Reconstruction Analysis and Leave-One-Group-Out Extrapolation

Figure 6 shows held-out CFD fields, surrogate predictions, and absolute error maps (seed 2026, labeled by nMAE_c rank) for showcase cases on ρ (`DeepONet`, rank 15/42; 3-seed rank 3/42), T (`PINN continuity`, rank 17/42; 3-seed rank 1/42), and v (`FiLM-MLP`, rank 10/42; 3-seed rank 3/42). Supplementary Figure S1 shows the corresponding **median (rank 21/42)** and **worst (rank 42/42)** held-out cases per field:

Table 4: Three-seed (2026, 2027, 2028) out-of-fold accuracy across the 42 sCO_2 cases. For each target, the leading cross-configuration pooled model is listed first, followed by the best within-group mesh-aligned model where it achieves lower error (and the lowest-MAE pooled model on ρ).

Target (unit)	Model	Scope	Macro MAE	Mean-base MAE	Reduction (%)	Macro RMSE	Median nMAE ($\pm$SD)	Pooled R^2
ρ (kg m^{-3})	DeepONet	Pooled	35.1	131.6	73.3%	37.1	0.837 ± 0.160	0.862
	Concat-MLP	Pooled	26.3	131.6	80.0%	29.1	1.011 ± 0.069	0.936
p_{gauge} (kPa)	Extra Trees	Pooled	83.2	254.0	67.3%	83.7	2.91†	0.815
	POD-RBF	Aligned	90.1	254.0	64.5%	90.2	0.0483†	0.571
	Nearest case	Aligned	108.1	254.0	57.4%	108.2	0.0122†	0.776
T (°C)	PINN continuity	Pooled	5.48	13.71	60.0%	7.32	1.350 ± 0.044	0.736
	POD-RBF	Aligned	6.11	13.71	55.4%	7.42	0.974 ± 0.041	0.620
ε (m^2 s^{-3})	FiLM-MLP	Pooled	0.0978	0.1945	49.7%	0.315	0.192 ± 0.020	0.569
	POD-RBF	Aligned	0.127	0.1945	34.8%	0.369	0.177 ± 0.018	-0.575
k (10^{-4} m^2 s^{-2})	FiLM-MLP	Pooled	0.947	3.635	73.9%	1.57	0.242 ± 0.051	0.898
	POD-RBF	Aligned	2.00	3.635	45.0%	2.78	0.237 ± 0.015	-0.070
u (m s^{-1})	FiLM-MLP	Pooled	0.0381	0.1763	78.4%	0.0569	0.162 ± 0.015	0.801
v (m s^{-1})	FiLM-MLP	Pooled	0.0383	0.1972	80.6%	0.0591	0.175 ± 0.004	0.861
w (10^{-3} m s^{-1})	FiLM-MLP	Pooled	1.43	2.660	46.3%	3.46	0.272 ± 0.007	0.418
	POD-RBF	Aligned	1.01	2.660	62.1%	2.55	0.207 ± 0.008	0.650
	Nearest case	Aligned	1.08	2.660	59.3%	2.93	0.185 ± 0.009	0.551

† For gauge pressure, the Median nMAE column reports the centered spatial residual $\text{nRMSE}_{\text{res}}$ (1 kPa floor; Table S8).

median cases accurately capture loop-scale stratification and riser/downcomer velocity cores with errors localized near the cooler wall and elbows, whereas rank-42/42 cases occur at extreme boundary conditions (HHHC_P2000_90_bar_0.05kgs_T_315 on ρ and T, the sole case with $T_{\text{in}} = 315$ K and $\bar{T}_c = 47.5$°C) or in the near-stagnant HHVC_P20_90_bar_0.05kgs_T_305 run on v ($\sigma_c(v) = 1.5$ mm s^{-1}).

Table 5 summarizes the leave-one-group-out extrapolation study across the four structural groups (16 pooled models × 8 fields = 128 model–field evaluations per group; full metrics in Supplementary Table S9). When VHVC/small ($n = 4$) is held out, both the small-loop geometry (0.894×1.000 m, 90 bar, 305 K) and the VHVC orientation have been seen during training, allowing XGBoost and FiLM-MLP to maintain moderate median nMAE (ρ: 0.980, T: 1.609, ε: 0.268, k: 0.391, v: 0.686). Conversely, holding out VHVC/large ($n = 13$) or HHHC/small ($n = 20$) requires extrapolating across an unseen loop scale (3.010 × 4.080 m, $D_i = 13.84$ mm, $T_{\text{in}} = 283.5$ K) or an unseen top-horizontal cooler configuration ($Q = 1000$–2000 W), inflating the best density median nMAE to 3.854 and 4.601.

4.4 Limitations

Three limitations of this study should be noted: (1) because only final converged CFD-Post exports are available without solver convergence histories, we cannot separate a physical flow-regime transition from convergence to an alternative steady numerical branch in the two near-stagnant cases HHVC_P20_90_bar_0.05kgs_T_305 ($Q = 20$ W) and HHVC_P30_90_bar_0.05kgs_T_305 ($Q = 30$ W), nor can we verify the specific k–ε sub-variant, wall-function treatment, or unexported hydrostatic reference density ρ_0; (2) plane-weighted mean CO_2 density $\bar{\rho}_{\text{plane},c}$ varies across steady-state cases at the same $(p_{\text{op}}, T_{\text{in}})$ (Section 2.2), so the cases represent independent steady runs rather than a strictly constant-mass trajectory; and (3) seeds 2026, 2027, and 2028 re-randomize the five-fold stratified case splits in addition to model ini-

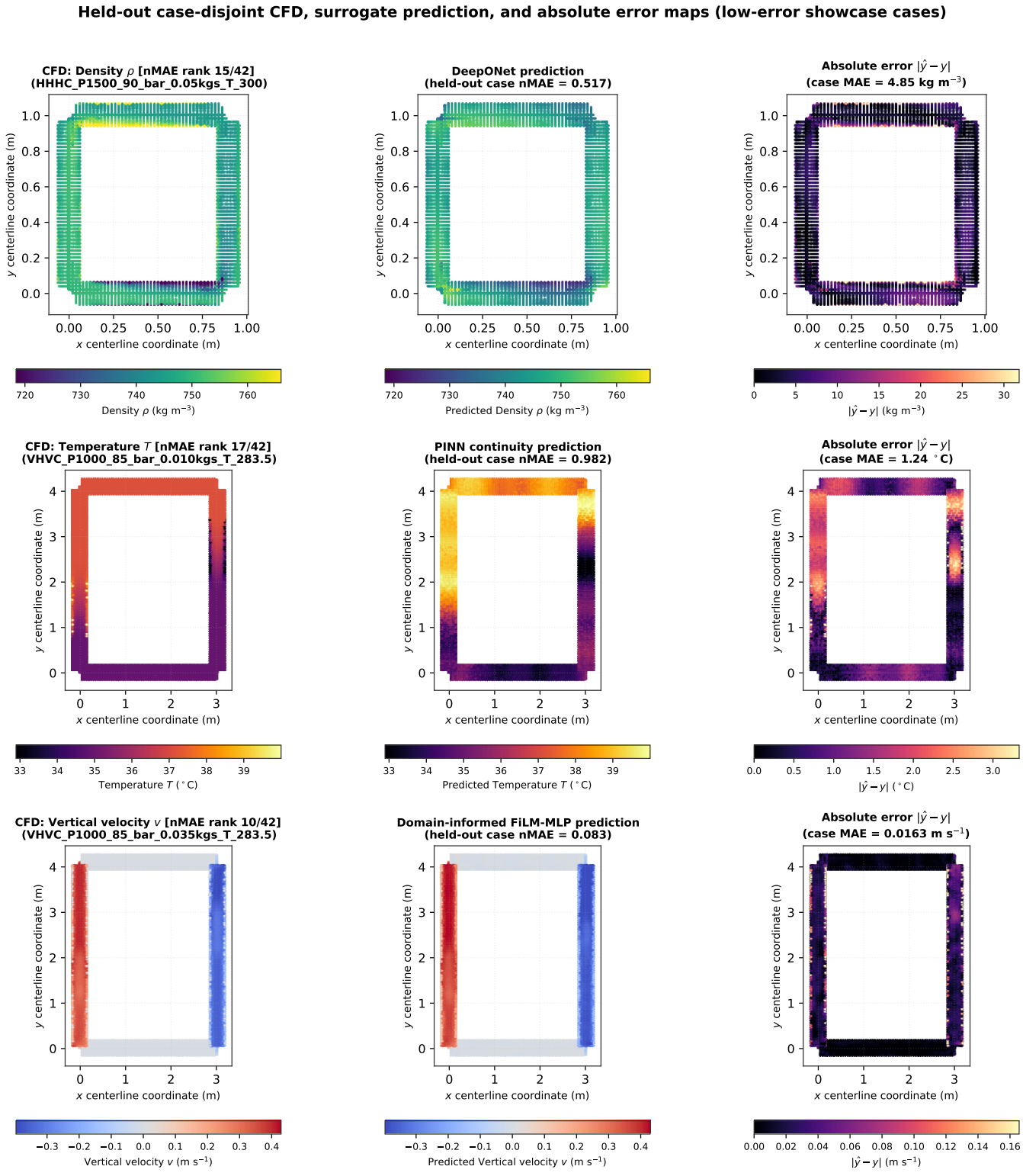


Figure 6: Held-out out-of-fold spatial reconstructions (seed 2026) comparing reference CFD (left), surrogate prediction (middle), and absolute error $|\hat{y} - y|$ (right) for ρ (`DeepONet`, rank 15/42), T (`PINN continuity`, rank 17/42), and v (`Domain-informed FiLM-MLP`, rank 10/42). Median (rank 21/42) and worst (rank 42/42) cases per field are shown in Supplementary Figure S1.

Table 5: Leave-one-group-out extrapolation summary across the four structural groups. Each cell reports the winning pooled model and its three-seed mean median case nMAE across the 16 models × 8 fields = 128 evaluations per group (p_{gauge} is reported via level MAE and residual $\text{nRMSE}_{\text{res}}$).

Field	`HHHC/small` ($n = 20$)	`HHVC/small` ($n = 5$)	`VHVC/small` ($n = 4$)	`VHVC/large` ($n = 13$)
ρ (nMAE)	`PINN cont.` (4.601)	`PINN data` (21.13)	`XGBoost` (0.980)	`Random Forest` (3.854)
T (nMAE)	`PINN cont.` (9.572)	`PINN data` (11.53)	`XGBoost` (1.609)	`PINN data` (1.682)
ε (nMAE)	`Transolver` (0.464)	`Concat-MLP` (0.822)	`FiLM-MLP` (0.268)	`GAMS-GNN` (0.258)
k (nMAE)	`Concat-MLP` (1.198)	`Concat-MLP` (1.242)	`FiLM-MLP` (0.391)	`GAMS-GNN` (0.752)
u (nMAE)	`Extra Trees` (0.710)	`Mean base.` (0.869)	`Mean base.` (0.875)	`Concat-MLP` (0.810)
v (nMAE)	`XGBoost` (0.733)	`Mean base.` (0.873)	`XGBoost` (0.686)	`Concat-MLP` (0.766)
w (nMAE)	`Transolver` (0.687)	`Mean base.` (0.736)	`GAMS-GNN` (0.692)	`Mean base.` (0.558)
p_{gauge} level MAE (kPa)	`DeepONet` (254.0)	`Mean base.` (536.2)	`F-FNO` (226.4)	`Ridge` (204.2)
p_{gauge} res. $\text{nRMSE}_{\text{res}}$	`Mean base.` (0.273)	`Mean base.` (0.197)	`Mean base.` (0.172)	`Extra Trees` (0.734)

tialization, and the small group sizes in `HHVC/small` ($n = 5$) and `VHVC/small` ($n = 4$) limit statistical power under group holdout.

5 Conclusions

We have presented a case-disjoint surrogate benchmark across 42 steady-state sCO_2 natural-circulation loop CFD simulations ($5,451,341$ primary sCO_2 nodes). Our main conclusions are:

1. **Case-disjoint validation and two-part pressure evaluation are essential:** Across the four nonlinear tree baselines, 80/20 random node splits understate five-fold case-disjoint case-macro MAE by 6.78 × –10.37× on density, 11.29 × –87.54× on gauge pressure, and 5.62 × –10.24× on temperature (3.34 × –4.05× and 2.95 × –4.39× in median case nMAE on ρ and T), while `Mean` and `Ridge` controls remain near 1.0 × –1.26×. Separating gauge pressure into its case-level mean offset and centered spatial residual prevents between-case pressurization shifts from obscuring within-loop gradient accuracy.

2. **Different physical fields favor distinct inductive biases:** Under three-seed five-fold case-disjoint cross-validation, multiplicative coordinate modulation (`Domain-informed FiLM-MLP`) leads all pooled models across all three velocity components (nMAE = $0.162, 0.175, 0.272$) and both turbulence fields (nMAE = 0.192 on ε, 0.242 on k); `DeepONet` (nMAE = 0.837) and `PINN continuity` (nMAE = 1.350) lead among pooled models on density and temperature; and within-group `POD-RBF` achieves the lowest within-case nMAE on temperature (0.974), ε (0.177), and k (0.237) when interpolating on a fixed mesh, whereas uniform voxelized spectral operators (`F-FNO`, `U-FNO`) underperform on slender pipe geometries ($D_i/L < 1.7\%$).

3. **Cross-geometry extrapolation requires multi-scale coverage:** Leave-one-group-out tests (128 model–field evaluations per group) show that pooled models transfer across heater orientations when the loop scale is covered in training (`VHVC/small`), but degrade when extrapolating across a 3.4 × –4.1× change in loop dimensions and operating regime (`VHVC/large`, `HHHC/small`).

Data and Code Availability

All preprocessing, benchmark training, statistical testing, and plotting scripts, along with the three-seed metric tables, pressure/density audit CSV, and supplementary material (Section S1, Figure S1, and Tables S0a–S10), are publicly available at https://github.com/Barghav777/SCWR_3D.

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